Faculty of Science and Information Technology

Outcome Based Curriculum and Syllabus

4-year B.Sc. Program in Computer Science and Engineering

March 2018

Daffodil International University

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Table of Contents

1.0	*Introduction*		
	1.1	Preamble	3
	1.2	Vision	3
	1.3	Mission	3
	1.4	Eligibility for Admission	3
	1.5	Credit Requirement and Duration of the Program	3
2.0	*Evaluation and Grading*		
	2.1	The Grading System	3
	2.2	Grading and Performance Evaluation	4
	2.3	Evaluation of Thesis/Project/Internship Work	5
3.0	*Outcome Based Education Component*		
	3.1	The Graduate Attributes	5
	3.2	Program Educational Objectives	6
	3.3	The Program Outcome	6
	3.4	Mapping PEO and PO with GA	7
4.0	*Program Detail*		
	4.1	Type of Courses	7
	4.2	Grouping of Courses	8
	4.3	Suggested Outline for Execution	8
	4.4	Courses Detail	14
	4.5	Elective Courses	54

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1.0 Introduction

1.1 Preamble

The B.Sc. Program in Computer Science & Engineering (CSE) is designed with the concept of real-life project based and outcome based education (OBE) model to produce skilled graduates in the field of ICT to satisfy the growing demands of computer science and engineering graduates both at home and abroad. It provides the students an huge opportunity to obtain broad knowledge of Computer Science, Computer Engineering with freedom to tailor the program according to the student's individual needs.

1.2 Vision

"Be the forerunner towards global knowledge hub with striving for the pursuit of excellence in CSE education, research, entrepreneurship and technological services to the society"

1.3 Mission

Department of Computer Science and Engineering is committed to:

a. Achieve academic and research excellence in CSE program through innovative teaching and learning with ICT based pedagogy
b. Prepare and groom students in professional form to face challenges of the ICT industry
c. Promote research and collaborations among the faculty and the students to create state of the art research environment
d. Promote quality and ethics among the students
e. Motivate students to acquire entrepreneur skills to become global leader

1.4 Eligibility for Admission

Students having minimum 2.5 GPA or second division both in SSC and HSC from science background may apply for admission into CSE as per the guideline of University Grants Commission (UGC) of Bangladesh. Students completing five O-level subjects and at least two A-level subjects and obtaining at least GPA 2.0 may apply for admission.

1.5 Credit Requirement and Duration of the Program

In order to obtain B.Sc. in Computer Science and Engineering, students will have to complete 148 credits with at least CGPA 2.5. If any student fails in any course, she/he will get opportunity to improve the grade by retaking the same in the subsequent semesters. The program having 12 semesters normally extends over four academic years.

Students willing to obtain a Bachelor of Science in Computer Science and Engineering will have to follow the general guidelines of degree requirements of the university.

2.0 Evaluation and Grading

2.1 The Grading System

The performance of a student in a given course/module is based on a scheme of continuous assessment. For theory courses this continuous assessment is made through a set of class tests, class evaluation, class participation, homework assignment, presentation, mid semester and a semester final examinations. The assessment in laboratory/practical courses is made through observation of the student at work during the class vivavoce during laboratory hours, lab tests and lab final exam.

Each course has a certain number of credits, which describes, its corresponding weights. A letter grade is awarded in relation to grade point. A students performance is measured both by the number of credits completed satisfactorily and by the weighted average of the grade point earned. A minimum grade point average (GPA) is essential for satisfactory progress. A minimum number of earned credits also have to be acquired in order to qualify for the degree. Letter grades and corresponding grade points will be awarded in accordance with the provisions shown below in Table 1:

Table 1: UGC Grading System

Letter Grade	Grade Point	Numerical Grade
A+	4.00	80% and above
A	3.75	75% to less than 80%
A-	3.50	70% to less than 75%
B+	3.25	65% to less than 70%
B	3.00	60% to less than 65%
B-	2.75	55% to less than 60%
C+	2.50	50% to less than 55%
C	2.25	45% to less than 50%
D	2.00	40% to less than 45%
F	0.00	less than 40%

Subject in which the student gets F grades shall not be counted towards credit requirements and for the calculation of Grade Point Average (GPA).

2.2 Grading and Performance Evaluation

Final grade in each course will be given on the basis of performance in class attendance, in-course examinations, assignments, midterm test and final examination as shown in Table 2:

Table 2: Theory class assessment

Type	Percentage
Class Attendance	7%
In-course Tests	15%
Assignment	5%
Class presentation	8%
Midterm Test	25%
Semester Final Examination	40%
Total	**100%**

Final grade in each laboratory course will be given on the basis of class attendance, lab performance, lab report and lab final examination as indicated below in Table 3:

Table 3: Laboratory class assessment

Type	Percentage
Class Attendance	10%
Performance	25%
Lab Report	25%
Lab Final Examination	40%
Total	**100%**

2.3 Evaluation of Thesis/Project/Internship Work

Students in their final year of the study enroll Thesis/Project/Internship. In order to complete the the work, this is given in two phases, Phase-I and Phase-II, in the last two semesters. In order to ensure the quality of the work, students are offered to join at Level-3 Term-3 as Pre-Phase I and proceed to attend the title defense. In the proceeding semesters, based on follow-ups and completion of the Pre-Phase II, Phase-1 and Phase-2, student proceed to qualify the pre-defense to be eligible for the final defense.

For the evaluation of the project work and grading, the assessment shown in Table 4 is followed.

Table 4: Final year project assessment

Type	Percentage
Project Defense	50%
Project Report	50%
Total	**100%**

A panel of examiners including the Head of the department, two internal members and one external member will conduct the project defense and also examine the project report.

3.0 Outcome Based Education (OBE) Components

3.1 The Graduate Attributes (GA)

The Graduate Attributes (GA) are the knowledge skills and attitudes which the students have at the time of graduation. These attributes are generic and are common to all engineering programs. These Graduate Attributes are identified by ABET accreditation.

GA1. Engineering Knowledge: Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.

GA2.Problem Analysis: Identify, formulate, research literature, and analyze complex Engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.

GA3. Design/Development of Solutions: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.

GA4.Conduct Investigations of Complex Problems: Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.

GA5. Modern Tool Usage: Create, select, and apply appropriate techniques, resource s, and modern engineering and IT tools including prediction and modelling to complex engineering activities with an understanding of the limitations.

GA6. The Engineer and Society: Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.

GA7. Environment and Sustainability: Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.

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GA8. Ethics: Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.
GA9. Individual and Team Work: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.
GA10. Communication: Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.
GA11. Project Management and Finance: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.
GA12.Life-Long Learning: Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

3.2 The Program Educational Objectives (PEO)

The Programme Educational Objectives (PEOs) are defined and developed based on the interactions with various stakeholders. Their interests, social relevance and contributions are taken in to account in defining and developing the PEOs. The program educational objectives are as follows in Table 5:

Table 5: List of PEOs

PEO1	Apply computer science and engineering theory blended with mathematics and modern tools and techniques to solve real-life problems.
PEO2	Conceptualize, Design, Implement, Test and Maintain software and hardware systems based on requirement specifications.
PEO3	Communicate effectively with team members, engage in applying technologies and lead teams in industry for information systems.
PEO4	Evaluate and assess the computing systems from the view point of quality, security, privacy, cost, utility, etiquette and ethics.
PEO5	Engage in lifelong learning, career enhancement and adapt to changing professional and societal needs.

3.3 The Program Outcome (PO)

Program learning outcomes are statements that describe what learners will know and be able to do when they graduate from a program. The program outcome are as follows:

PO 1: Be able to acquire and apply knowledge for real-life problem solving
PO 2: Be able to identify problems and formulate solutions
PO 3: Be able to analyze and design computing systems
PO 4: Be able to conduct, evaluate, and interpret computing experiments
PO 5: Be able to demonstrate team work and leadership
PO 6: Be able to communicate and collaborate
PO 7: Be able to understand the impact of engineering decisions
PO 8: Have an understanding on professional and ethical responsibility
PO 9: Apply lifelong learning
PO 10: Be able to contribute beyond professional careers

3.4 Mapping PEO and PO with GA

The Programme Educational Objectives (PEO) and Program Outcome (PO) are mapped against Graduate Attribute (GA) as follows: (3: most likely; 2: likely; 1: partial)

Mapping of program educational objectives with graduate attributes

PEO	GA1	GA2	GA3	GA4	GA5	GA6	GA7	GA8	GA9	GA10	GA11	GA12
PEO1	3	3	3	3	1	1	1	-	-	-	2	2
PEO2	3	3	3	3	3	1	1	-	2	2	3	1
PEO3	2	2	2	2	2	1	2	-	3	3	3	2
PEO4	2	2	2	3	2	2	1	3	-	1	2	1
PEO5	1	1	1	1	2	3	2	3	2	2	1	3

Mapping of program outcomes with graduate attributes

PO	GA1	GA2	GA3	GA4	GA5	GA6	GA7	GA8	GA9	GA10	GA11	GA12
PO1	3	3	3	1	1	-	-	-	2	-	-	-
PO2	3	3	-	-	-	2	1	-	2	-	3	-
PO3	3	3	3	3	3	3	3	-	2	-	2	2
PO4	3	1	2	2	2	-	-	-	1	-	-	2
PO5	3	3	2	3	2	-	-	-	-	-	-	-
PO6	2	1	3	1	-	-	-	2	-	-	-	-
PO7	-	-	-	-	-	2	-	3	-	3	-	2
PO8	-	-	-	2	2	-	-	-	-	-	-	3
PO9	-	-	-	-	2	-	3	-	3	3	2	-
PO10	-	-	-	-	-	3	3	-	-	-	-	3

Mapping of program outcomes with program educational objectives

PEO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10
PEO1	3	2	2	2	2	2	1	1	1	1
PEO2	3	3	3	3	2	2	1	2	2	2
PEO3	2	3	3	3	3	3	3	2	3	1
PEO4	3	3	3	2	3	3	3	3	2	1
PEO5	1	2	2	2	3	2	3	3	3	3

4.0 Program Detail

4.1 Types of Courses

In Computer Science and Engineering (CSE) program, there are different types of courses, namely Humanities and Business courses, Mathematics and General Science courses, Core courses and Elective courses. Humanities courses are useful for the students to grow as a good citizen with social values and norms; mathematics and general science courses are necessary for the background preparations; core courses form the nucleus of the bachelor degree program, and elective courses provide in-depth understanding in selected areas. Elective courses are offered from a list of elective courses in the final year of the program.

4.2 Grouping of Courses

Students willing to obtain a bachelor's degree in Computer Science and Engineering (CSE) will have to follow the general guidelines of degree requirements of the university. The courses organized under four groups are shown in the following table.

Courses Summary:

Year	Lab Credits	Core Credits	Math Credits	GED Credits	Others/Proj
1	10	11	06	17	0
2	15	13	08	04	0
3	17	16	00	00	0
4	06	19	00	00	06
Total Credits	**48**	**59**	**14**	**21**	**06**

Grand Total: 48+59+14+21+06 = 148 credits

4.3 Suggested Outline for Execution

The courses are offered as shown in the following tables. (L=Lecture; T=Tutorial; P: Practical)

Foundation Year

Level-1 Term-1

SL	Code	Course Title	L	T	P	Credits
1	CSE112	Computer Fundamentals	1.5	0	1.5	3
2	MAT111	Basic Mathematics	3	0	0	3
3	ENG113	Basic Functional English and English Spoken	3	0	0	3
4	PHY113	Basic Physics	3	0	0	3
5	PHY114	Basic Physics Lab	0	0	2	1
6	GED111	History of Bangladesh and Bangla Language	3	0	0	3
		Total	**13.5**	**0**	**3.5**	**16**

Level-1 Term-2

SL	Code	Course Title	L	T	P	Credits
1	MAT121	Mathematics-II: Calculus, Complex Variables and Linear Algebra	3	1	0	3
2	CSE122	Programming and Problem Solving	3	1	0	2
3	CSE123	Problem Solving Lab	1	0	3	2
4	ENG123	Writing and Comprehension	3	0	0	3
5	CSE124	Business Application Design	0	0	2	1
6	GED121	Bangladesh Studies	3	0	0	3
		Total	**13**	**2**	**5**	**14**

Level-1 Term-3

SL	Code	Course Title	L	T	P	Credits
1	CSE131	Discrete Mathematics	3	0	0	3
2	CSE132	Electrical Circuits	3	0	0	1
3	CSE133	Electrical Circuits Lab	0	0	3	2
4	CSE134	Data Structure	3	1	0	2
5	CSE135	Data Structure Lab	0	0	3	2
6	CSE136	Software Project I	0	0	2	1
7	GED131	Art of Living	3	0	0	3
		Total	**12**	**1**	**8**	**14**

Table 6: Mapping of CSE Courses with Program Outcomes (Foundation)

Course	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10
CSE112	√	√			√					
CSE122	√	√								
CSE123	√	√	√		√					
CSE124	√	√								
CSE134	√	√								
CSE135	√	√	√		√					
CSE136	√	√	√		√					

Sophomore

Level-2 Term-1

SL	Code	Course Title	L	T	P	Credits
1	MAT211	Engineering Mathematics	3	0	0	3
2	CSE212	Basic Electronics	1.5	0	0	1
3	CSE213	Basic Electronics Lab	0	0	3	2
4	CSE214	Algorithms	3	1	0	2
5	CSE215	Algorithms Lab	0	0	3	2
6	CSE216	Software Project II	0	0	2	1
7	ACT211	Financial and Managerial Accounting	3	0	0	2
		Total	**10.5**	**1**	**8**	**13**

Level-2 Term-2

SL	Code	Course Title	L	T	P	Credits
1	CSE221	Object Oriented Programming	3	0	0	2
2	CSE222	Object Oriented Programming Lab	0	1	2	2
3	STA221	Statistics and Probability	0	0	3	3
4	CSE223	Digital Electronics	1.5	0	0	1
5	CSE224	Digital Electronics Lab	0	0	3	2
6	CSE225	Data Communication	3	0	0	3
7	CSE226	Software Project III	0	0	2	1
		Total	**7.5**	**1**	**10**	**14**

Level-10 Term-3

SL	Code	Course Title	L	T	P	Credits
1	CSE231	Microprocessor, Embedded Systems and IoT	1.5	0	0	1
2	CSE232	Microprocessor, Embedded Systems and IoT Lab	0	1	3	2
3	CSE233	Object Oriented Programming II	1.5	0	0	1
4	CSE234	Object Oriented Programming II Lab	0	1	3	2
5	CSE235	Numerical Methods	3	0	0	3
6	CSE236	Math for Computer Science	3	0	0	2
7	CSE237	Software Project IV	0	0	2	1
		Total	**9**	**2**	**8**	**12**

Table 7: Mapping of CSE Courses with Program Outcomes (Sophomore)

Course	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10
CSE214	√	√	√			√				
CSE215	√	√	√	√	√	√				
CSE216	√	√	√	√	√	√				
CSE221	√	√	√							
CSE222	√	√	√	√	√	√				
CSE226	√	√	√	√	√	√				
CSE231	√	√	√							
CSE232	√	√	√	√	√	√				
CSE233	√	√	√		√					
CSE234	√	√	√	√	√	√				
CSE237	√	√	√	√	√	√				

Junior

Level-3 Term-1

SL	Code	Course Title	L	T	P	Credits
1	CSE311	Database Management System	3	0	0	2
2	CSE312	Database Management System Lab	0	1	3	2
3	CSE313	Computer Networks	1.5	0	0	1
4	CSE314	Computer Networks Lab	0	1	3	2
5	CSE315	Artificial Intelligence	1.5	0	0	1
6	CSE316	Artificial Intelligence Lab	0	1	3	2
7	CSE317	Software Project V	0	0	2	1
		Total	**6**	**3**	**11**	**11**

Level-3 Term-2

SL	Code	Course Title	L	T	P	Credits
1	CSE321	Data Mining and Machine Learning	3	0	0	2
2	CSE322	Data Mining and Machine Learning Lab	0	1	3	2
3	CSE323	Operating Systems	1.5	0	0	1
4	CSE324	Operating Systems Lab	0	1	3	2
5	CSE325	System Analysis and Design	3	0	3	3
6	ECO321	Economics	3	0	0	2
7	CSE326	Research and Innovation	1.5	0	0	1
		Total	**12**	**2**	**9**	**13**

Level-3 Term-3

SL	Code	Course Title	L	T	P	Credits
1	CSE331	Complier Design	3	0	0	2
2	CSE332	Complier Design Lab	0	1	3	2
3	CSE333	Software Engineering	1.5	0	3	3
4	CSE334	Pervasive Computing	1.5	0	0	1
5	CSE335	Pervasive Computing and Mobile App Development Lab	0	0	3	2
6	CSE336	Software Project VI	0	0	3	1
		Total	**6**	**1**	**12**	**11**

Table 8: Mapping of CSE Courses with Program Outcomes (Junior)

Course	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10
CSE311	√	√	√							
CSE312	√	√	√	√	√	√				
CSE313	√	√	√					√		
CSE314	√	√	√	√	√	√				
CSE315	√	√	√							
CSE316	√	√	√	√	√	√				
CSE317	√	√	√	√	√	√				
CSE321	√	√	√							
CSE322	√	√	√							
CSE323	√	√	√	√	√	√				
CSE324	√	√	√							
CSE325	√	√		√	√	√				
CSE326	√			√	√	√				
CSE331	√	√	√							
CSE332	√	√	√	√	√	√				
CSE333	√							√	√	
CSE334	√		√	√						
CSE335	√	√	√	√	√	√				
CSE336	√	√	√	√	√	√				

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Senior

Level-4 Term-1

SL	Code	Course Title	L	T	P	Credits
1	CSE411	Computer Architecture and Organization	3	0	0	3
2	CSE412	Big Data and IoT	1.5	0	0	1
3	CSE413	Big Data and IoT Lab	0	1	3	2
4	CSE414	Web Engineering	1.5	0	0	1
5	CSE415	Web Engineering Lab	0	0	3	2
		Total	**6**	**1**	**12**	**09**

Level-4 Term-2

SL	Code	Course Title	L	T	P	Credits
1	CSE421	Computer Graphics	1.5	0	0	1
2	CSE422	Computer Graphics Lab	0	0	3	2
3	CSE423	Information Security	3	0	0	3
4	CSE XXX	Elective I	3	0	0	3
5	CSE499	Project / Internship (Phase I, to be completed in Level-4 Term-3	0	0	0	3
		Total	**7.5**	**0**	**3**	**12**

Level-4 Term-3

SL	Code	Course Title	L	T	P	Credits
1	CSE498	Social and Professional Issues in Computing	3	0	0	3
2	CSE XXX	Elective II	3	0	0	3
3	CSE499	Project / Internship (Phase II, continued from Level 4 Term 2)	3	0	0	3
		Total	**9**	**0**	**0**	**9**

Table 9: Mapping of CSE Courses with Program Outcomes (Senior)

Course	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10
CSE411	√	√	√							
CSE412	√	√	√							
CSE413	√	√	√	√	√	√				
CSE414	√	√	√							
CSE415	√	√	√	√	√	√				
CSE416	√	√	√					√	√	√
CSE421	√	√	√							
CSE422	√	√	√	√	√	√				
CSE423	√	√	√							
CSE498	√	√	√		√			√	√	√

Summary:

Based on above execution plan, the summary of credits for B.Sc. in CSE program is shown below:

Year/Level	Term 1	Term 2	Term 3	Total
1	16 Credits	14 Credits	14 Credits	44 Credits
2	15 Credits	14 Credits	12 Credits	41 Credits
3	11 Credits	11 Credits	11 Credits	33 Credits
4	09 Credits	12 Credits	9 Credits	30 Credits
Total				**148 Credits**

Courses Summary:

Year	Lab Credits	Core Credits	Math Credits	GED Credits	Others/Proj
1	10	11	06	17	0
2	15	13	08	04	0
3	17	16	00	00	0
4	06	19	00	00	06
Total Credits	**48**	**59**	**14**	**21**	**06**

Grand Total: 48+59+14+21+06 = 148 credits